CLASS 10 · CBSE · SAMPLE PAPER SOLUTIONS

MathSetu Sample Paper 1: Mathematics (Standard)

Step-by-step solutions with the marks for each question

The question numbers match the question paper.

1 Section A (20 questions of 1 mark each)

1. The HCF of $2^3 \times 3^2 \times 5$ and $2^2 \times 3^3 \times 7$ is: (a) 12 (b) 36 (c) 72 (d) 108 1 mark

SOLUTION

Take the smallest power of each common prime: $2^2 \times 3^2 = 36$.

Answer: (b) 36

2. If α and β are the zeroes of $x^2 - 7x + 10$, then $\frac{1}{\alpha} + \frac{1}{\beta}$ equals: (a) $\frac{7}{10}$ (b) $\frac{10}{7}$ (c) $-\frac{7}{10}$ (d) $\frac{3}{10}$ 1 mark

SOLUTION

$\alpha + \beta = 7$ and $\alpha\beta = 10$, so $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{7}{10}$.

Answer: (a) $\frac{7}{10}$

3. A quadratic polynomial whose zeroes are 3 and -4 is: (a) $x^2 - x - 12$ (b) $x^2 + x + 12$ (c) $x^2 + x - 12$ (d) $x^2 - 7x + 12$ 1 mark

SOLUTION

Sum of zeroes $= -1$ and product $= -12$, so the polynomial is $x^2 - (-1)x + (-12) = x^2 + x - 12$.

Answer: (c) $x^2 + x - 12$

4. The pair of equations $3x + ky = 5$ and $6x + 8y = 7$ has no solution when k equals: (a) 2 (b) 4 (c) 8 (d) 16 1 mark

SOLUTION

For no solution, $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$: $\frac{3}{6} = \frac{k}{8}$ gives $k = 4$, and $\frac{5}{7} \neq \frac{1}{2}$.

Answer: (b) 4

5. The roots of $2x^2 - 5x + 3 = 0$ are: (a) real and equal (b) real and distinct (c) not real (d) both negative 1 mark

SOLUTION

$D = b^2 - 4ac = 25 - 24 = 1 > 0$, so the roots are real and distinct (they are 1 and $\frac{3}{2}$).

Answer: (b) real and distinct

6. What is the 15th term of the AP $8, 14, 20, \ldots$? (a) 86 (b) 92 (c) 98 (d) 104 1 mark

SOLUTION

$a = 8$, $d = 6$: $a_{15} = 8 + 14 \times 6 = 92$.

Answer: (b) 92

7. Find the sum of the first 10 terms of the AP $3, 7, 11, \ldots$: (a) 190 (b) 200 (c) 210 (d) 220 1 mark

SOLUTION

$S_{10} = \frac{10}{2}\left[2 \times 3 + 9 \times 4\right] = 5 \times 42 = 210$.

Answer: (c) 210

8. The distance between the points $(-3, 4)$ and $(5, -2)$ is: (a) 6 units (b) 8 units (c) 10 units (d) 14 units 1 mark

SOLUTION

Distance $= \sqrt{(5+3)^2 + (-2-4)^2} = \sqrt{64 + 36} = 10$ units.

Answer: (c) 10 units

9. The mid-point of the line segment joining $(2, -5)$ and $(-6, 9)$ is: (a) $(-2, 2)$ (b) $(-4, 4)$ (c) $(4, -7)$ (d) $(2, -2)$ 1 mark

SOLUTION

Mid-point $= \left(\frac{2-6}{2}, \frac{-5+9}{2}\right) = (-2, 2)$.

Answer: (a) $(-2, 2)$

10. In $\triangle ABC$, D lies on AB and E lies on AC with $DE \parallel BC$. If $AD = 3$ cm, $DB = 4.5$ cm and $AE = 4$ cm, then EC is: (a) 5 cm (b) 5.5 cm (c) 6 cm (d) 7.5 cm 1 mark

SOLUTION

By the Basic Proportionality Theorem, $\frac{AD}{DB} = \frac{AE}{EC}$, so $\frac{3}{4.5} = \frac{4}{EC}$ and $EC = 6$ cm.

Answer: (c) 6 cm

11. Two similar triangles have corresponding sides in the ratio $3:5$. If the perimeter of the smaller triangle is 27 cm, the perimeter of the larger triangle is: (a) 36 cm (b) 40 cm (c) 45 cm (d) 75 cm 1 mark

SOLUTION

Perimeters of similar triangles are in the ratio of the corresponding sides: $\frac{27}{P} = \frac{3}{5}$, so $P = 45$ cm.

Answer: (c) 45 cm

12. In $\triangle ABC \sim \triangle PQR$, $\angle A = 50^\circ$ and $\angle Q = 60^\circ$. Then $\angle C$ is: (a) 50° (b) 60° (c) 70° (d) 80° 1 mark

SOLUTION

Corresponding angles are equal, so $\angle B = \angle Q = 60^\circ$. Then $\angle C = 180^\circ - 50^\circ - 60^\circ = 70^\circ$.

Answer: (c) 70°

13. A point is 13 cm from the centre of a circle of radius 5 cm. The length of the tangent from the point to the circle is: (a) 8 cm (b) 12 cm (c) 14 cm (d) 18 cm 1 mark

SOLUTION

The radius is perpendicular to the tangent, so length $= \sqrt{13^2 - 5^2} = \sqrt{144} = 12$ cm.

Answer: (b) 12 cm

14. PA and PB are tangents to a circle with centre O. If $\angle APB = 70^\circ$, then $\angle AOB$ is: (a) 70° (b) 90° (c) 110° (d) 140° 1 mark

SOLUTION

In quadrilateral $OAPB$, $\angle OAP = \angle OBP = 90^\circ$, so $\angle AOB = 180^\circ - 70^\circ = 110^\circ$.

Answer: (c) 110°

15. If $\sin\theta = \frac{3}{5}$ and θ is acute, then $\tan\theta$ is: (a) $\frac{4}{5}$ (b) $\frac{3}{4}$ (c) $\frac{4}{3}$ (d) $\frac{5}{3}$ 1 mark

SOLUTION

$\cos\theta = \sqrt{1 - \frac{9}{25}} = \frac{4}{5}$, so $\tan\theta = \frac{3/5}{4/5} = \frac{3}{4}$.

Answer: (b) $\frac{3}{4}$

16. The value of $2\tan^2 45^\circ + \cos^2 30^\circ - \sin^2 60^\circ$ is: (a) 0 (b) 1 (c) 2 (d) 3 1 mark

SOLUTION

$2(1)^2 + \frac{3}{4} - \frac{3}{4} = 2$.

Answer: (c) 2

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17. The area of a sector of a circle of radius 6 cm with central angle 60° is: (a) 3π cm^2 (b) 6π cm^2 (c) 9π cm^2 (d) 12π cm^2 1 mark

SOLUTION

Area $= \dfrac{60}{360} \times \pi \times 6^2 = 6\pi$ cm^2.

Answer: (b) 6π cm^2

18. For a distribution, the mean is 20 and the median is 22. Using the empirical relation, the mode is: (a) 18 (b) 24 (c) 26 (d) 28 1 mark

SOLUTION

Mode $= 3\,\text{Median} - 2\,\text{Mean} = 66 - 40 = 26$.

Answer: (c) 26

19. **Assertion (A):** When a die is thrown once, the probability of getting a prime number is $\frac{1}{2}$. **Reason (R):** The prime numbers on a die are 2, 3 and 5. Choose the correct option: (a) Both A and R are true, and R is the correct explanation of A. (b) Both A and R are true, but R is not the correct explanation of A. (c) A is true, but R is false. (d) A is false, but R is true. 1 mark

SOLUTION

There are 3 primes (2, 3, 5) among 6 equally likely outcomes, so the probability is $\frac{3}{6} = \frac{1}{2}$. Both statements are true, and R is the reason for A.

Answer: (a)

20. **Assertion (A):** Two cubes, each of edge 4 cm, are joined end to end. The surface area of the cuboid formed is 192 cm^2. **Reason (R):** When two solids are joined, the surface area of the new solid is less than the sum of their surface areas. Choose the correct option from the four options given in Question 19. 1 mark

SOLUTION

The cuboid measures $8 \times 4 \times 4$ cm, so its surface area is $2(32 + 32 + 16) = 160$ cm^2, not 192 cm^2. A is false.

The two faces that are joined are hidden, so R is true.

Answer: (d)

2 Section B (5 questions of 2 marks each)

21. Two alarm bells ring at intervals of 18 minutes and 24 minutes. They ring together at 8:00 a.m. At what time will they next ring together? 2 marks

SOLUTION

They ring together again after the LCM of 18 and 24 minutes.

$18 = 2 \times 3^2$ and $24 = 2^3 \times 3$, so LCM $= 2^3 \times 3^2 = 72$ minutes $= 1$ hour 12 minutes.

Answer: 9:12 a.m.

22. Find the value of k for which $kx^2 - 12x + 9 = 0$ has two equal roots. **OR** Solve for x and y: $2x + 3y = 13$ and $5x - 2y = 4$. 2 marks

SOLUTION

Equal roots need $D = 0$: $(-12)^2 - 4(k)(9) = 0$, so $144 = 36k$ and $k = 4$.

OR: Multiply the first equation by 2 and the second by 3: $4x + 6y = 26$ and $15x - 6y = 12$. Adding, $19x = 38$, so $x = 2$.

Then $2(2) + 3y = 13$ gives $y = 3$.

Answer: $k = 4$; **or** $x = 2$, $y = 3$

23. In $\triangle ABC$, points D and E lie on AB and AC with $AD = 4$ cm, $DB = 6$ cm, $AE = 5$ cm and $EC = 7.5$ cm. Show that $DE \parallel BC$. If $BC = 15$ cm, find DE. 2 marks

SOLUTION

$\frac{AD}{DB} = \frac{4}{6} = \frac{2}{3}$ and $\frac{AE}{EC} = \frac{5}{7.5} = \frac{2}{3}$. The ratios are equal, so by the converse of the Basic Proportionality Theorem, $DE \parallel BC$.

Then $\triangle ADE \sim \triangle ABC$ (AA), so $\frac{DE}{BC} = \frac{AD}{AB} = \frac{4}{10}$ and $DE = \frac{4}{10} \times 15 = 6$ cm.

Answer: $DE = 6$ **cm**

24. Evaluate $4\sin^2 60^\circ + 3\tan^2 30^\circ - 8\sin 45^\circ \cos 45^\circ$. **OR** If $\sin(A + B) = 1$ and $\cos(A - B) = \frac{\sqrt{3}}{2}$, where $0^\circ < A + B \le 90^\circ$ and $A > B$, find A and B. 2 marks

SOLUTION

$4 \times \frac{3}{4} + 3 \times \frac{1}{3} - 8 \times \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} = 3 + 1 - 4 = 0$.

OR: $\sin(A + B) = 1$ gives $A + B = 90^\circ$, and $\cos(A - B) = \frac{\sqrt{3}}{2}$ gives $A - B = 30^\circ$. Adding, $2A = 120^\circ$.

Answer: 0; or $A = 60^\circ$, $B = 30^\circ$

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25. A bag has 5 red, 8 white and 7 green balls. One ball is drawn at random. Find the probability that the ball is (i) white (ii) not red. 2 marks

SOLUTION

Total balls $= 5 + 8 + 7 = 20$.

(i) $P(\text{white}) = \frac{8}{20} = \frac{2}{5}$ (ii) $P(\text{not red}) = \frac{8+7}{20} = \frac{3}{4}$

Answer: (i) $\frac{2}{5}$ (ii) $\frac{3}{4}$

3 Section C (6 questions of 3 marks each)

26. Prove that $\sqrt{5}$ is an irrational number. 3 marks

SOLUTION

Suppose $\sqrt{5}$ is rational. Then $\sqrt{5} = \frac{a}{b}$, where a and b are co-prime positive integers.

Squaring, $a^2 = 5b^2$. So 5 divides a^2, and therefore 5 divides a. Write $a = 5c$.

Then $25c^2 = 5b^2$, so $b^2 = 5c^2$. So 5 divides b^2, and therefore 5 divides b.

Now 5 divides both a and b, which contradicts the fact that they are co-prime.

So the assumption is wrong, and $\sqrt{5}$ is irrational.

27. Find the zeroes of $6x^2 - 7x - 3$ and verify the relationship between the zeroes and the coefficients. **OR** The sum of the digits of a two-digit number is 9. The number obtained by reversing the digits is 27 more than the original number. Find the number. 3 marks

SOLUTION

$6x^2 - 7x - 3 = 6x^2 - 9x + 2x - 3 = 3x(2x - 3) + 1(2x - 3) = (3x + 1)(2x - 3)$. The zeroes are $-\frac{1}{3}$ and $\frac{3}{2}$.

Sum of zeroes $= -\frac{1}{3} + \frac{3}{2} = \frac{7}{6} = -\frac{b}{a}$. Product of zeroes $= -\frac{1}{3} \times \frac{3}{2} = -\frac{1}{2} = \frac{-3}{6} = \frac{c}{a}$. Verified.

OR: Let the tens digit be x and the units digit be y. Then $x + y = 9$, and $(10y + x) - (10x + y) = 27$ gives $y - x = 3$.

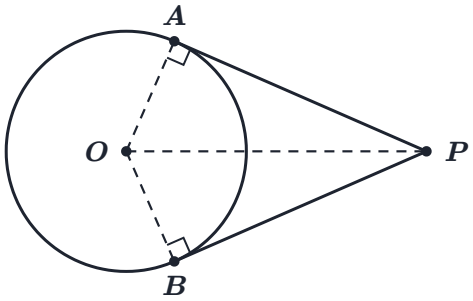
Adding the two equations, $2y = 12$, so $y = 6$ and $x = 3$.

Answer: zeroes $-\frac{1}{3}$ and $\frac{3}{2}$; or the number is 36

28. Show that the two tangent segments drawn to a circle from a point outside it are equal in length. 3 marks

SOLUTION

Let PA and PB be the tangents from an external point P to a circle with centre O. Join OA, OB and OP.



In $\triangle OAP$ and $\triangle OBP$: $\angle OAP = \angle OBP = 90^\circ$ (a radius is perpendicular to the tangent at the point of contact), $OA = OB$ (radii) and $OP = OP$ (common).

So $\triangle OAP \cong \triangle OBP$ (RHS).

Therefore $PA = PB$ (CPCT).

29. Prove that $\dfrac{\sin\theta}{1+\cos\theta} + \dfrac{1+\cos\theta}{\sin\theta} = 2\,\mathrm{cosec}\,\theta$. 3 marks

SOLUTION

$$\text{LHS} = \frac{\sin^2\theta + (1+\cos\theta)^2}{\sin\theta\,(1+\cos\theta)} = \frac{\sin^2\theta + 1 + 2\cos\theta + \cos^2\theta}{\sin\theta\,(1+\cos\theta)}$$

$$= \frac{2 + 2\cos\theta}{\sin\theta\,(1+\cos\theta)}, \text{ using } \sin^2\theta + \cos^2\theta = 1$$

$$= \frac{2(1+\cos\theta)}{\sin\theta\,(1+\cos\theta)} = \frac{2}{\sin\theta}$$

$$= 2\,\mathrm{cosec}\,\theta = \text{RHS. Proved.}$$

30. In a circle of radius 14 cm, a chord subtends a right angle at the centre. Find the area of the minor segment it cuts off. **OR** The minute hand of a clock is 10.5 cm long. Find the area it sweeps in 20 minutes. 3 marks

SOLUTION

Area of the sector $= \frac{90}{360} \times \frac{22}{7} \times 14 \times 14 = 154$ cm^2.

Area of the right triangle formed by the two radii $= \frac{1}{2} \times 14 \times 14 = 98$ cm^2.

Area of the minor segment $= 154 - 98 = 56$ cm^2.

OR: In 20 minutes the minute hand turns through $\frac{20}{60} \times 360^\circ = 120^\circ$. Area swept $= \frac{120}{360} \times \frac{22}{7} \times 10.5 \times 10.5 = \frac{1}{3} \times 346.5 = 115.5$ cm^2.

Answer: 56 cm^2; or 115.5 cm^2

31. Find the mean of the following distribution of marks. 3 marks

Marks	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50
Number of students	3	5	12	6	4

SOLUTION

Use the class marks x_i: 5, 15, 25, 35, 45.

Marks	f_i	x_i	$f_i x_i$
0 – 10	3	5	15
10 – 20	5	15	75
20 – 30	12	25	300
30 – 40	6	35	210
40 – 50	4	45	180
Total	30		780

Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{780}{30} = 26$.

Answer: mean $= 26$ marks

4 Section D (4 questions of 5 marks each)

32. A bus covers 300 km at a uniform speed. If its speed had been 10 km/h more, the journey would have taken 1 hour less. Find the speed of the bus. **OR** In an AP, the 4th term is 11 and the sum of the first 10 terms is 155. Find the AP, its 20th term and the sum of its first 20 terms. 5 marks

SOLUTION

Let the speed be x km/h. Then $\frac{300}{x} - \frac{300}{x+10} = 1$.

$300(x+10) - 300x = x(x+10)$, so $x^2 + 10x - 3000 = 0$.

$(x+60)(x-50) = 0$, so $x = 50$ or $x = -60$. A speed cannot be negative, so $x = 50$.

Check: $\frac{300}{50} = 6$ h and $\frac{300}{60} = 5$ h, a difference of 1 hour.

OR: $a + 3d = 11$, and $\frac{10}{2}(2a + 9d) = 155$ gives $2a + 9d = 31$.

Put $a = 11 - 3d$: $22 - 6d + 9d = 31$, so $d = 3$ and $a = 2$. The AP is $2, 5, 8, 11, \ldots$

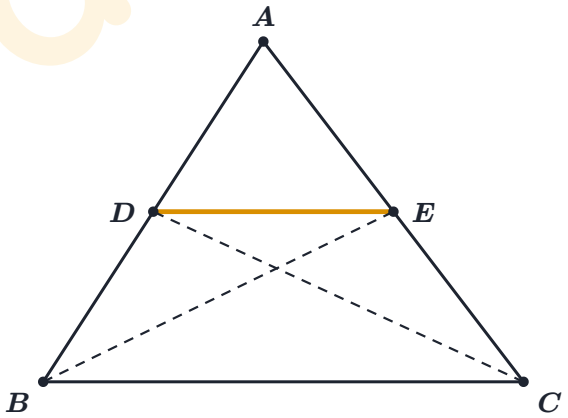
$a_{20} = 2 + 19 \times 3 = 59$ and $S_{20} = \frac{20}{2}(2 + 59) = 610$.

Answer: 50 km/h; or AP $2, 5, 8, \ldots$ with $a_{20} = 59$ and $S_{20} = 610$

33. Prove that if a line is drawn parallel to one side of a triangle to cut the other two sides in distinct points, then it divides those two sides in the same ratio. Use this result to find x if, in $\triangle PQR$, $ST \parallel QR$ with S on PQ and T on PR, and $PS = x$, $SQ = x - 2$, $PT = x + 2$, $TR = x - 1$. 5 marks

SOLUTION

In $\triangle ABC$, let $DE \parallel BC$ with D on AB and E on AC. Join BE and CD. Draw $EN \perp AB$ and $DM \perp AC$.



$\frac{\text{ar}(ADE)}{\text{ar}(BDE)} = \frac{\frac{1}{2} \times AD \times EN}{\frac{1}{2} \times DB \times EN} = \frac{AD}{DB}$ and $\frac{\text{ar}(ADE)}{\text{ar}(DEC)} = \frac{\frac{1}{2} \times AE \times DM}{\frac{1}{2} \times EC \times DM} = \frac{AE}{EC}$.

$\triangle BDE$ and $\triangle DEC$ stand on the same base DE and lie between the same parallels DE and BC, so $\text{ar}(BDE) = \text{ar}(DEC)$.

Therefore $\frac{AD}{DB} = \frac{AE}{EC}$. Proved.

In $\triangle PQR$: $\frac{PS}{SQ} = \frac{PT}{TR}$, so $\frac{x}{x-2} = \frac{x+2}{x-1}$. Then $x^2 - x = x^2 - 4$, which gives $x = 4$.

Answer: $x = 4$

34. From the top of a tower, the angles of depression of the top and the bottom of a pole 10 m high are 30° and 60°. Find the height of the tower and its distance from the pole. **OR** Two ships are on opposite sides of a lighthouse 75 m high, in line with its foot. From the top of the lighthouse their angles of depression are 30° and 45°. Find the distance between the ships. (Take $\sqrt{3} = 1.732$.) 5 marks

SOLUTION

Let the tower be h m high and d m from the pole. The angles of depression equal the angles of elevation from the pole.

From the bottom of the pole: $\tan 60^\circ = \dfrac{h}{d}$, so $h = \sqrt{3}\,d$.

From the top of the pole: $\tan 30^\circ = \dfrac{h-10}{d}$, so $h - 10 = \dfrac{d}{\sqrt{3}}$.

Then $\sqrt{3}\,d - 10 = \dfrac{d}{\sqrt{3}}$, so $3d - 10\sqrt{3} = d$ and $d = 5\sqrt{3} \approx 8.66$ m. Hence $h = \sqrt{3} \times 5\sqrt{3} = 15$ m.

OR: For the ship seen at 45°: distance from the foot $= \dfrac{75}{\tan 45^\circ} = 75$ m. For the ship seen at 30°: distance $= \dfrac{75}{\tan 30^\circ} = 75\sqrt{3}$ m.

Distance between the ships $= 75 + 75\sqrt{3} = 75(1 + 1.732) = 204.9$ m.

Answer: height 15 m, distance $5\sqrt{3} \approx 8.66$ m; or 204.9 m

35. A solid toy is a cone standing on a hemisphere of the same radius, 7 cm. The total height of the toy is 31 cm. Find its volume and its total surface area. **OR** A solid metal cone of radius 6 cm and height 24 cm is melted and recast into solid spheres of radius 2 cm. Find the number of spheres, and the surface area of one sphere. (Take $\pi = 3.14$.) 5 marks

SOLUTION

Height of the cone $= 31 - 7 = 24$ cm, so its slant height is $l = \sqrt{7^2 + 24^2} = 25$ cm.

Volume $= \dfrac{2}{3}\pi r^3 + \dfrac{1}{3}\pi r^2 h = \dfrac{1}{3}\pi r^2(2r + h) = \dfrac{1}{3} \times \dfrac{22}{7} \times 49 \times 38 = \dfrac{5852}{3} \approx 1950.67$ cm^3.

Total surface area $= 2\pi r^2 + \pi r l = \pi r(2r + l) = \dfrac{22}{7} \times 7 \times 39 = 858$ cm^2.

OR: Volume of the cone $= \dfrac{1}{3}\pi \times 6^2 \times 24 = 288\pi$ cm^3. Volume of one sphere $= \dfrac{4}{3}\pi \times 2^3 = \dfrac{32\pi}{3}$ cm^3.

Number of spheres $= 288\pi \div \dfrac{32\pi}{3} = 27$. Surface area of one sphere $= 4\pi r^2 = 4 \times 3.14 \times 4 = 50.24$ cm^2.

Answer: volume ≈ 1950.67 cm^3 and surface area 858 cm^2; or 27 spheres, each of surface area 50.24 cm^2

5 Section E (3 case study questions of 4 marks each)

36. A student starts a fitness plan. She does 12 push-ups on the first day and, on every following day, 3 more than on the day before. (i) How many push-ups does she do on the 10th day? (ii) On which day does she do 60 push-ups? (iii) Find the total number of push-ups in the first 15 days, **OR** find the number of days in which the total reaches 255. 4 marks, case study

SOLUTION

The daily numbers form an AP with $a = 12$ and $d = 3$.

(i) $a_{10} = 12 + 9 \times 3 = 39$.

(ii) $12 + (n-1) \times 3 = 60$ gives $n - 1 = 16$, so $n = 17$.

(iii) $S_{15} = \frac{15}{2}[24 + 14 \times 3] = \frac{15}{2} \times 66 = 495$.

OR: $\frac{n}{2}[24 + 3(n-1)] = 255$ gives $3n^2 + 21n - 510 = 0$, that is $n^2 + 7n - 170 = 0$. So $(n-10)(n+17) = 0$ and $n = 10$.

Answer: (i) 39 (ii) the 17th day (iii) 495; or 10 days

37. Three lamp posts in a park stand at the points $P(2, 1)$, $Q(8, 3)$ and $R(6, 9)$ on a grid whose unit is 1 metre. (i) Find the distance PQ. (ii) Find the mid-point of QR. (iii) Find the point that divides PR in the ratio $1 : 3$, **OR** show that $\triangle PQR$ is an isosceles right triangle. 4 marks, case study

SOLUTION

(i) $PQ = \sqrt{(8-2)^2 + (3-1)^2} = \sqrt{40} = 2\sqrt{10}$ m.

(ii) Mid-point of $QR = \left(\frac{8+6}{2}, \frac{3+9}{2}\right) = (7, 6)$.

(iii) The point is $\left(\frac{1 \times 6 + 3 \times 2}{1+3}, \frac{1 \times 9 + 3 \times 1}{1+3}\right) = (3, 3)$.

OR: $PQ^2 = 40$, $QR^2 = (6-8)^2 + (9-3)^2 = 40$ and $PR^2 = (6-2)^2 + (9-1)^2 = 80$. So $PQ = QR$ and $PQ^2 + QR^2 = PR^2$: the triangle is isosceles and right-angled at Q.

Answer: (i) $2\sqrt{10}$ m (ii) $(7, 6)$ (iii) $(3, 3)$; or shown

38. A housing society recorded the water used in one day by each of its 40 households, as shown in the table. (i) Write the modal class. (ii) How many households used less than 200 litres? (iii) Find the median of the data, **OR** find its mode. 4 marks, case study

Water used (litres)	100 – 150	150 – 200	200 – 250	250 – 300	300 – 350
Number of households	5	8	13	9	5

SOLUTION

(i) The greatest frequency is 13, so the modal class is 200 – 250.

(ii) $5 + 8 = 13$ households.

(iii) The cumulative frequencies are 5, 13, 26, 35, 40 and $\frac{N}{2} = 20$, so the median class is 200 – 250, with $l = 200$, $cf = 13$, $f = 13$, $h = 50$.

Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 200 + \frac{20 - 13}{13} \times 50 \approx 200 + 26.92 = 226.92$ litres.

OR: Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 200 + \frac{13 - 8}{26 - 8 - 9} \times 50 = 200 + \frac{250}{9} \approx 227.78$ litres.

Answer: (i) 200 – 250 (ii) 13 (iii) median ≈ 226.92 litres; or mode ≈ 227.78 litres

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